Diffusion pseudotime robustly reconstructs lineage branching

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The temporal order of differentiating cells is intrinsically encoded in their single-cell expression profiles. We describe an efficient way to robustly estimate this order according to diffusion pseudotime (DPT), which measures transitions between cells using diffusion-like random walks. Our DPT software implementations make it possible to reconstruct the developmental progression of cells and identify transient or metastable states, branching decisions and differentiation endpoints.

Gene expression profiling can provide important insights into how developmental processes are regulated. However, the inherent stochasticity and external influences that impinge on genetic regulation often lead to strong heterogeneity and asynchrony in the timing of cellular programs. Time-resolved bulk transcriptomics averages over these effects and obscures the underlying gene dynamics. Single-cell profiling techniques, in contrast, allow a systematic observation of a cell's regulatory state[1] as they capture cells at various developmental stages[2,3]. Since cells are destroyed during measurement, gene dynamics and hence the sequence of cellular programs have to be inferred from static snapshot data. This is generally achieved by ordering cells according to expression similarity, which is known as pseudotemporal ordering. Initially proposed for bulk expression[4], pseudotemporal ordering was later extended to single-cell data, for example, RNA-seq and mass cytometry data[5–7]. However, existing pseudotime algorithms face challenges regarding robustness and scalability when applied to data with branching lineages. These difficulties are amplified by the advent of high-throughput experimental techniques such as Drop-seq[8,9] and MARS-seq[10], which can profile tens of thousands of cells, albeit at high noise rates.

To overcome these problems, we introduce a pseudotime measure we call diffusion pseudotime (DPT). DPT is a random-walk-based distance that is computed based on simple Euclidian distances in the 'diffusion map space'. The diffusion map is a nonlinear method for recovering the low-dimensional structure underlying high-dimensional observations[11]. It organizes data by defining coordinates as dominant eigenvectors of a transition matrix T that describes random walks between data points—in this case, between cells in distinct stages of the differentiation process. A diffusion map strongly reduces noise and is able to represent branching data, but it has only been used for visualization thus far[12,13]. Our main contribution is the derivation of a measure on this space (DPT) that is suitable for recovering the dynamics of biological processes underlying the data, in particular, developmental trajectories from single-cell data. DPT (see Online Methods, equation (1)) is defined as the ordering of cells by comparing their probabilities of differentiating toward different cell fates.

To compute DPT from single-cell expression data (**Fig. 1a**), we first built a transition matrix by convolving Gaussians centered at nearby cells, thereby effectively constructing a weighted nearest-neighbor graph of the data. For each cell, we then determined the probabilities of transitioning to each other cell in the data set using random walks of any length on this graph; these walks can be seen as a proxy for the cells' probabilities of differentiating toward different fates. The probabilities for each cell are stored in a vector, and the DPT between two cells is calculated as the Euclidian distance between their two vectors (**Supplementary Note 1**). We measured the developmental progression of each cell in the data set by computing its DPT with respect to a specified root cell.

DPT benefits from the favorable properties of diffusion maps, such as robustness to noise and sampling density[11]. The latter facilitates the application of DPT to experiments in which the number of profiled cells varies between different developmental stages (see Online Methods). In addition, DPT involves closed-form mathematical expressions, rendering it computationally efficient and applicable to data sets comprising tens of thousands of cells. DPT does not rely on dimension reduction and can thus detect subtle changes in high-dimensional gene expression patterns.

DPT-based analysis then proceeds by identifying branching points that occur after cells progress from a root cell x through a common 'trunk trajectory' (**Fig. 1a**). Branching of the trunk is identified by measuring the correlation of two pseudotime sequences along trajectories that start from the root cell x and from a cell y with maximal DPT with respect to x. Whereas these sequences are anticorrelated on their connecting trajectory, they become correlated in a separate branch leading to a cell z. Branching points are thus identified by cells for which the two sequences switch from anticorrelated to correlated behavior (see Online Methods and **Supplementary Note 1**).

After branch identification, we proposed to determine metastable (e.g., quiescent) cells by density analysis (see Online Methods). We assumed that the speed at which cells progress through

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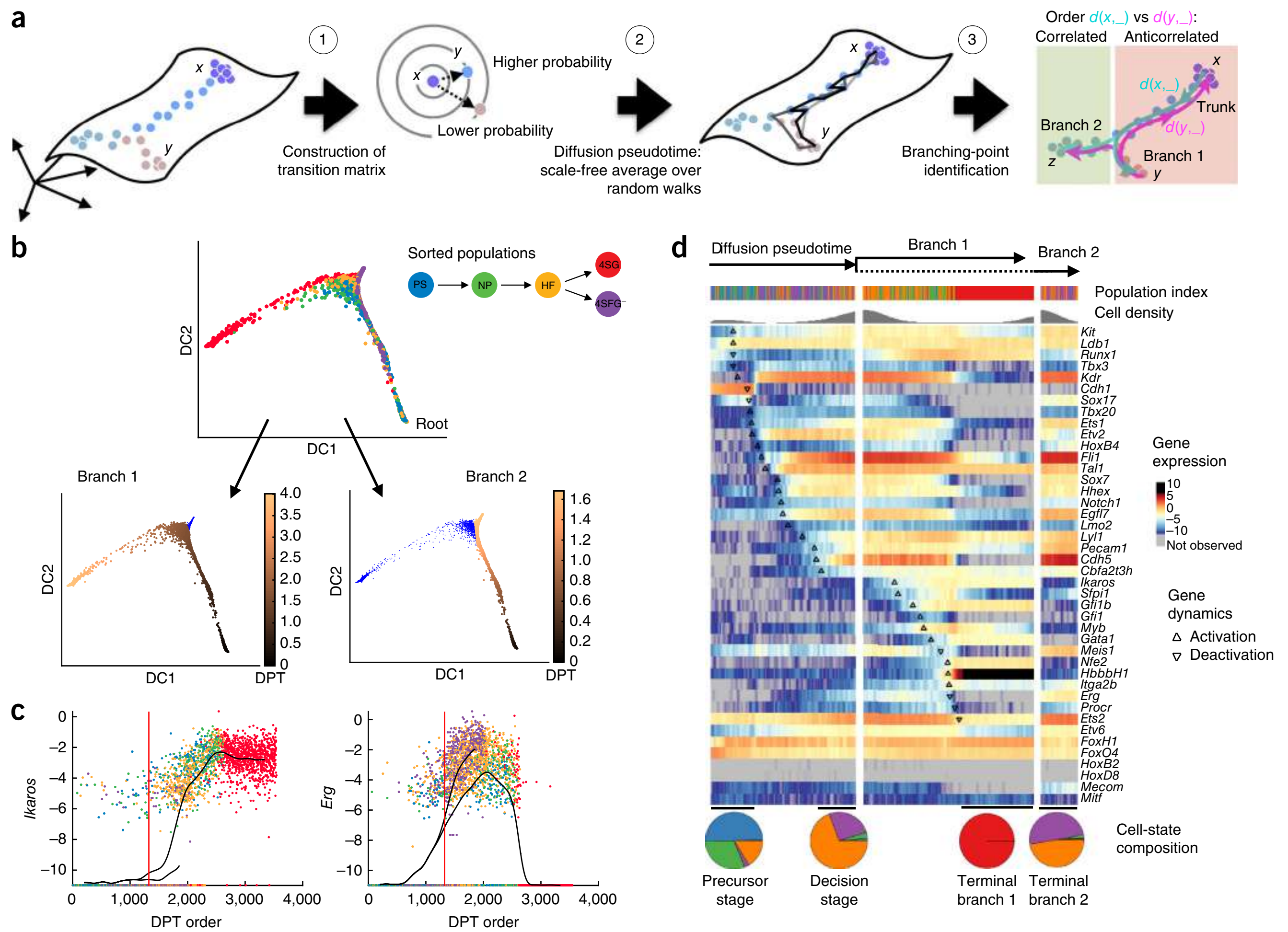

Figure 1 | Diffusion pseudotime reveals temporal ordering and cellular decisions on the single cell level. (**a**) The diffusion transition matrix T_{xy} is constructed by computing the overlap of local kernels at the expression levels of cells *x* and *y* (1). Diffusion pseudotime dpt*(x,y)* approximates the geodesic distance between *x* and *y* on the mapped manifold (2). Branching points are identified as points where anticorrelated distances from branch ends become correlated (3). (**b**) Application of DPT to single-cell qPCR of 42 genes in 3,934 single cells during early hematopoiesis[13], sorted from primitive streak (PS), neural plate (NP), head fold (HF), four somite GFP negative (4SG–) and four somite GFP positive (4SG+). DPT identifies the endothelial branch 1 (4SG$^-$) and the erythroid branch 2 (4SG$^+$) (blue cells in bottom graphs). (**c**) Dynamics of genes *Erg* and *Ikaros* in both branches. Black lines show the moving average over 50 adjacent cells. The red vertical line depicts the branching point. (**d**) Heatmap of gene expression (smoothed over 50 adjacent cells), with cells ordered by DPT and branching and genes ordered according to first major change (see **Supplementary Note 2**, section 2), which is indicated by black triangles (upward: activation, downward: deactivation). Pie charts (bottom) show the fraction of cells in the four metastable states (metastable state populations are high-density DPT regions indicated by the black horizontal line above the pie charts).

developmental states is inversely proportional to their density. Since cell accumulation is a sign of slow progression, metastable cells can be recognized by pseudotimes with high densities.

We performed a DPT analysis of single-cell qPCR data focusing on early blood development in mice[13]. Early hematopoietic cells branch to become either red blood cells or endothelial-like cells. DPT ordered cells along their developmental trajectories and identified two branches (**Fig. 1b**) which corresponded to the reported blood (branch 1) and endothelial branches (branch 2)[13]. Plotting gene expression versus pseudotime, we found patterns across developmental stages that are known to be characteristic of blood progenitors (**Fig. 1c,d**), namely the hemangioblast-like sequence[14] (subsequent upregulation of *Cdh1* to *Tal1* and *Cdh5*) in the trunk[13] and the endothelial differentiation route[13] in branch 2 (elevated levels of *Pecam1*, *Erg* and *Sox17*, among others). In branch 1, we found sequential expression of *Etv2*, *Tal1*, *Runx1* and *Gata1*[15], a sequence of gene activations characteristic for erythroid development. DPT further allowed us to distinguish early transitions (characterized by *Ikaros* expression, for example) from late transitions (e.g., *Erg* expression) (**Fig. 1c**) as well as a number of intermediate regulatory events[13] until the onset of *Hbb-bH1* expression (**Fig. 1d**). This information is crucial for the understanding of regulatory interactions; genes that undergo transitions (**Supplementary Note 2**) earlier than others are candidates for regulators of the differentiation process.

Metastable cells are identified by pseudotime regions of high density (**Fig. 1d** and **Supplementary Fig. 1**). We found four such states: precursor cells, hemangioblast-like cells at the decision state, erythroid-like and endothelial-like cells. Notably, both decision and precursor states consist of cell mixtures from two or three different stages, stressing the asynchrony of developmental stages that could not be resolved without pseudotemporal ordering.

To identify key decision genes, we quantified expression differences for DPT-inferred subgroups and experimentally sorted cells

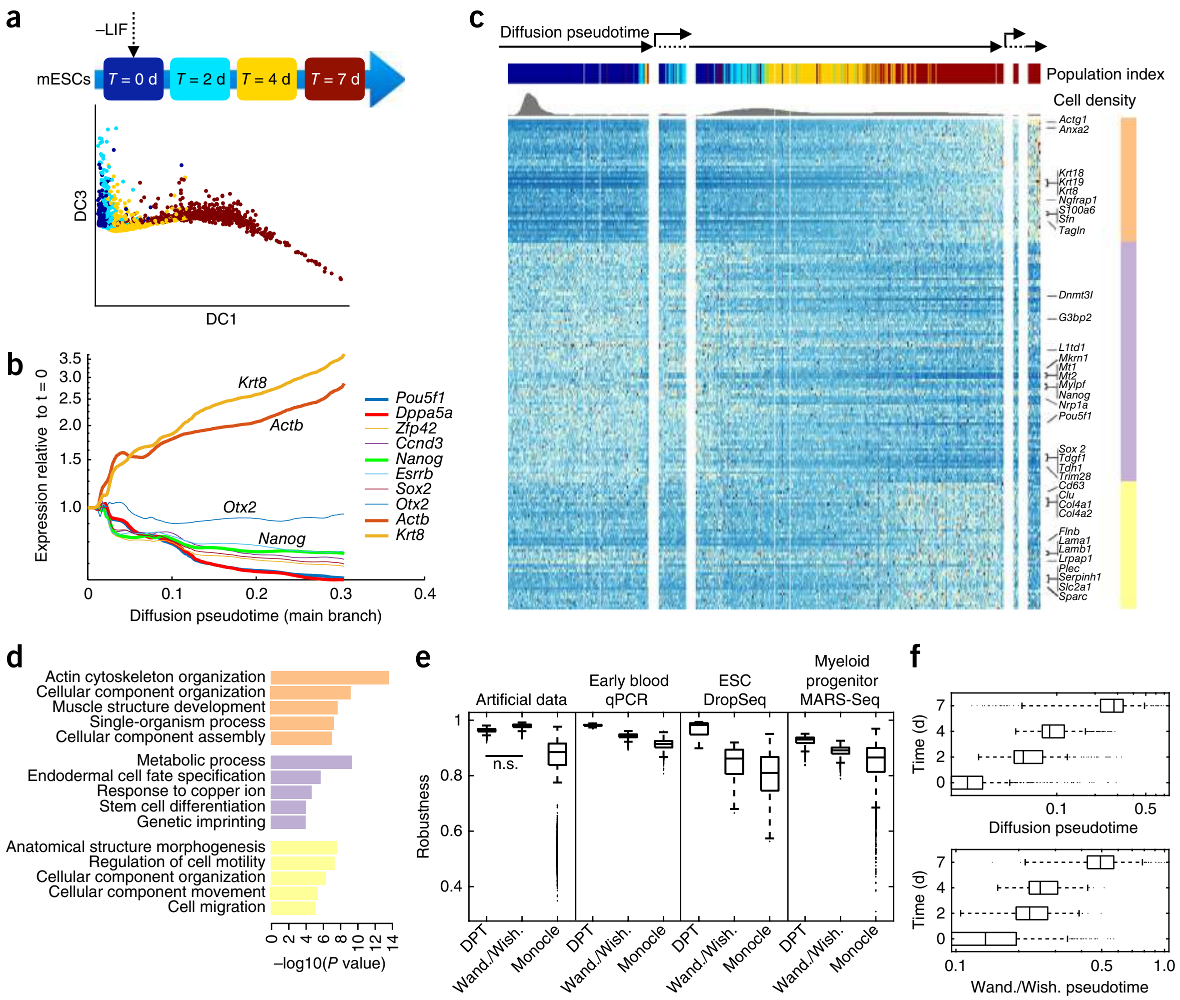

Figure 2 | Diffusion pseudotime identifies differentiation dynamics from scRNA-seq data[9]. (**a**) Mouse ESCs after LIF withdrawal were harvested at $T = 0$, 2, 4 and 7 d and profiled with the inDrop protocol, yielding 2,717 cells with 24,175 observed unique transcripts[9]. Visualization using diffusion maps shows temporal dynamics across the 7 d. (**b**) Pseudotemporal dynamics of the expression of selected genes. Compare with **Figure 7b** of Klein *et al.*[9]. (**c**) Heatmap of gene expression, with cells ordered by DPT and branches (separated by white vertical bars) and genes ordered according to hierarchical clustering. The heatmap depicts gene dynamics after hierarchical clustering (see **Supplementary Fig. 4b**); clusters indicated by color bar on right. (**d**) Gene ontology enrichment shows a cellular reorganization signature (orange), a metabolic signature for differentiation (purple) and a cell motility signature (yellow). (**e**) Comparisons of robustness of DPT, Wanderlust/Wishbone (Wand./Wish.) and Monocle by self-concordance measure on bootstrap samples for several data sets (**Supplementary Note 5**). DPT shows higher robustness (self concordance) across all data sets (all two-sided *t*-test significance levels $P <$ 0.001, except for the nonsignificant (n.s.) difference from Wishbone in the artificial data). (**f**) Kendal rank correlation of pseudotime with experimental days. DPT orders cells (see Online Methods) significantly better than pseudotemporal ordering by Wanderlust/Wishbone (Kendal rank correlation 0.77 ± 10^{-3} versus 0.70 ± 10^{-3}). (**e**,**f**) Boxes indicate median and first and third quartile, whiskers extend to ±1.5× the interquartile ratio divided by the square root of the number of observations, and single points denote measurements outside this range.

using MAST[16] (**Supplementary Fig. 2**). Testing differential gene expression for the 'decision' versus 'precursor' groups inferred by DPT resulted in 32 out of 42 significant genes, including *Cbfa2t3h* and *Pecam1*, which are known to indicate hematopoietic and endothelial development[14], respectively (**Supplementary Fig. 2a**). Comparing this with differentially expressed genes between the experimental groups head fold (HF) and primitive streak (PS) (**Supplementary Fig. 2b**) resulted in a less clear picture; in the latter case, the log-fold change was consistently lower than in the former case (**Supplementary Table 1**). Also, differential gene expression between HF and 4SG cells failed to identify endothelial differentiation but brought up erythroid factors (*Runx1*, *Ikaros* and *Gfi1b*, among others; see **Supplementary Fig. 2c–e** and **Supplementary Table 2**). In summary, when comparing differentially expressed genes between metastable states, DPT resolves the developmental patterns more clearly than the experimental sorting of cell populations (**Supplementary Fig. 2**). This suggests that heterogeneous marker expression across the sorted populations is more likely due to cells being in different developmental stages than to mere stochasticity of gene expression.

As a second demonstration, we applied DPT to single-cell RNA (scRNA)-seq data generated using droplet barcoding[9]. Klein *et al.*[9] monitored the transcriptomic profiles and heterogeneity in differentiation of mouse embryonic stem (ES) cells after leukemia inhibitory factor (LIF) withdrawal (**Fig. 2a**). After cell-cycle normalization (see Online Methods and **Supplementary Figs. 3,4**), DPT found a single differentiation path from which two small populations branched off. As a striking example, we can resolve upregulation of epiblast markers (*Krt8*/*18*/*19*) and downregulation of pluripotency factors (*Nanog*; **Fig. 2b**) at subday resolution in contrast with the original study (see Klein *et al.*[9], **Fig. 7b**). The first population of 151 cells that branched off (**Fig. 2c**) was enriched in apoptosis-related genes (e.g., *Clu*, *Cd63*; **Supplementary Fig. 5**). The second branching event gave rise to a population of 27 cells with increased primitive endoderm markers (e.g., *Serpinh1* and *Sparc*; **Fig. 2c**) and a population of 67 cells with increased epiblast markers (*Krt8* and *Actg1*, **Fig. 2c**; *Krt8* and *Actb*, **Fig. 2b**). Clustering of the gene expression dynamics (**Supplementary Note 3**) identified three major clusters, which we observed to differ both in their pseudotemporal behaviors (**Fig. 2c** and **Supplementary Fig. 4**) and biological functions (**Fig. 2d**). For example, one cluster consisted of pluripotency factors, which we found to be active in early pseudotime and then to decrease gradually. Altogether, DPT analysis leads to an accurate, high-resolution reconstruction of early embryonic stem cell differentiation events and transcription factor dynamics.

Finally, on an adult hematopoiesis scRNA-seq data set[10], DPT identified the dominant branching into different myeloid lineages. It also found a subpopulation of lymphoid outliers and a graded transition reflecting erythroid differentiation, which differed from previously observed cluster sequences[10] (**Supplementary Note 4**).

DPT overcomes the problems regarding robustness and scalability of previous algorithms[5–7] that prevent these algorithms from finding new biology in many relevant settings (**Fig. 2e**, see Online Methods, **Supplementary Note 5**), and it is more accurate (**Fig. 2f**). We also clarified the general relationship between pseudotime and actual time measurements (see Online methods and **Supplementary Note 6**). In the future, robust computation of pseudotimes could allow regulatory relationships to be inferred without perturbation experiments[13], and DPT could enable scaling such inference to genome-wide models. Recently, pseudotemporal ordering has been applied to cell morphology to identify cell-cycle states[17]. DPT of cell morphology features instead of RNA expression would allow the inclusion of branching, for example, to identify cells switching into a quiescent state. In summary, DPT provides a robust and scalable tool to infer cellular trajectories from snapshot data in high-dimensional single-cell expression profiles. We provide R and MATLAB implementations of DPT (**Supplementary Software** and http://www.helmholtz-muenchen.de/icb/dpt).

METHODS

Methods and any associated references are available in the online version of the paper.

Accession codes. Gene Expression Omnibus: early blood qPCR data[13], accession code GSE61470; mouse embryonic stem cells inDrop data[9], accession code GSE65525; mouse myeloid progenitors MARS-seq data[10], accession code GSE72857.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

L.H. developed the method and the computational tools, performed the analysis and wrote the paper and the supplement. M.B. contributed to the analysis and biological interpretation of results and wrote the supplement. F.A.W. helped interpret the results and write the supplement, and he wrote the paper. F.B. helped interpret the results. F.J.T. conceived and supervised the study, contributed to the method development and wrote the paper with help from all coauthors.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Details of the diffusion pseudotime analysis. *Overview of algorithmic steps in the diffusion pseudotime analysis.* (1) Initialization using the following user-provided parameters:

a. A data matrix of dimension number of cells times number of genes, b. the index or indices of one or several root cells and c. diffusion maps options 'classic' with a single parameter kernel width or 'locally scaled' with a single parameter number of nearest neighbors that we use for adjusting a local kernel width for each cell. (2) Computation of the transition matrix T. (3) Computation of the accumulated transition matrix M and of DPT with respect to the specified root cells. (4) Iterative assignment of cells to branches. (5) Identification of metastable states.

Definition of diffusion pseudotime. At the core of DPT is a transition matrix T that approximates the dynamic transitions of cells through stages of the differentiation process. This transition matrix is computed using a nearest neighbor graph whose edge weights have a Gaussian distribution with respect to Euclidian distance in gene expression space; transition probabilities correspond to edge weights. The eigenvectors of T are known as diffusion components[11] and have been used in diffusion maps for visualizing single-cell RNA-seq data[12,18]. While using only few diffusion components yields interpretable visualizations and amounts to a common dimensionality reduction method, important information may be lost by removing the remaining components. Consequently, DPT is based on the full-rank T rather than a low-rank approximation; i.e., we do not use diffusion maps as a dimensionality reduction method but as a method for representing and organizing the single-cell data.

We define diffusion pseudotime dpt(x,y), our novel distance measure, for two cells with index x and y as

$$\mathrm{dpt}(x, y) = \| \boldsymbol{M}(x,.) - \boldsymbol{M}(y,.) \|, \ \boldsymbol{M} = \sum_{t=1}^{\infty} \tilde{\boldsymbol{T}}^t. \tag{1}$$

Here, $\|\ldots\|$ denotes the L^2 norm. Instead of the probability $(T^t)_{xy}$ for a random walk of fixed length[11] t from x to y, in equation (1) we compute the accumulated transition probability $(M)_{xy}$ of visiting y when starting from x over random walks of all lengths t by summing over t. This is done using the modified transition matrix $\tilde{\boldsymbol{T}}$, which is defined as T without the first eigenspace. The first eigenspace can be associated with the steady state, and the transition matrix $\tilde{\boldsymbol{T}}$ can be thought of as encoding information about how this steady state is approached. We note that the main contribution to M is the pseudoinverse of T (its inverse being not defined), which is a standard object in the theory of Markov Processes but also in spectral clustering[19]. It is a strong computational advantage that M can be obtained in closed form; it reads

$$\boldsymbol{M} = \sum_{t=1}^{\infty} \tilde{\boldsymbol{T}}^t = (I - \tilde{\boldsymbol{T}})^{-1} - I \text{ where } \tilde{\boldsymbol{T}} = \boldsymbol{T} - \psi_0 \psi_0^T,$$

where ψ_0 denotes the eigenvector corresponding to the largest eigenvalue of T (see **Supplementary Note 1**). Fixing a known root cell x as start of the biological process of interest, the DPT of cell y, with respect to the root cell x, is dpt(x,y).

We point out that DPT is a measure of distance over random walks of arbitrary length, and can hence be considered 'scale-free' in contrast with diffusion distance as introduced by Coifman *et al.*[11]. Characterizations of distance using diffusion maps have mostly relied on diffusion distance, which involves as scale parameter t, the fixed length of random walks that occur in the definition of diffusion distance. In DPT, by summing over all random walk lengths, t is no longer present. DPT therefore has contributions from random walks of all scales. Further mathematical details on the definition of DPT are given in **Supplementary Note 1**.

Effects of sampling density. We point out that the DPT ordering of cells is (almost) independent of cell sampling density. Coifman *et al.*[11] show that this independence can be achieved by normalizing the transition matrix T with a proxy for sampling density. The result is that one can sample more (or less) cells in a specific region of the data manifold, but distances on the manifold, such as DPT or diffusion distance, do not change. We provide numerical evidence for this on simulated data (**Supplementary Note 7**, section 5). For this, we compute DPT using the original simulated data (related to a toggle switch), for which the sampling density versus DPT is strongly inhomogeneous; i.e., cells accumulate at certain DPT intervals (**Supplementary Note 7**, section 5). We then subsample from this original data set with the constraint of enforcing an (almost) homogeneous sampling density of cells with respect to DPT. The sampling density of this subsampled data differs strongly from the original data, whereas the DPT ordering stays almost the same (**Supplementary Note 7**, section 5).

Metastable states. Although DPT ordering is (almost) independent under sampling density changes, our definition of metastable states is not. As explained in the main text, our definition of metastable states is based on the assumption that sampling densities reflect how fast cells are passing through differentiation states. Hence, factors such as cell creation and death processes (e.g., when having a transit-amplifying set of cells) or a subjective choice of cells that are screened in the experiment influence our identification of metastable states.

Branching points. Branching points are determined by comparing two independent DPT orderings over cells, one starting at the root cell x and the other at its maximally distant cell y. The two sequences of pseudotimes are anticorrelated until the two orderings merge in a new branch, where they become correlated (**Fig. 1**). This criterion robustly identifies branching points, as we illustrate for simulation data for which the ground truth is known (**Supplementary Note 7**, section 1). One can repeat the procedure of branch finding iteratively in each of the branches that are found to identify further (i.e., more than three) sub-branches in the data (see **Supplementary Note 1** for further details).

Comparison of diffusion pseudotime to previous algorithms. *Numerical experiments.* When applying Monocle[5] and Wishbone[7] to the qPCR data from our first example, both fail to identify the endothelial branch (**Supplementary Note 7**, section 2). Similarly, for the scRNA-seq data from our second example, neither of the two methods identifies the important split between epiblast and primitive endoderm (**Supplementary Note 7**, section 3). Also for our third example, without extensive preprocessing only DPT identifies the dominant branching (**Supplementary Note 7**, section 4). A detailed simulation study on robustness (**Fig. 2e** and **Supplementary Note 5**) and comparisons on artificial data (**Supplementary Note 7**, section 1) further confirm the superior robustness of DPT. Finally, even when previous algorithms are able to obtain qualitatively meaningful results, we observe DPT to yield a higher quantitative accuracy of, for example, the ESC differentiation process in the scRNA-seq data from our second example in the main text; while the Kendall rank correlation of the DPT ordering with the true time ordering is $\hat{o} = 0.77 \pm 10^{-3}$, the Kendall rank correlation of Wanderlust/Wishbone is 0.70 ± 10^{-3} (see **Fig. 2f**). Kendall rank correlation ô is defined as:

$$\hat{o} = 2(m - m')/n(n-1)$$

where m is the number of concordant pair of cells (between pseudotime and experimental day), m' is the number of disconcordant pair of cells and n is the total number of cells.

Methodological comparison. The high robustness of DPT can theoretically be understood from its random-walk-based formulation. In comparison to Monocle, which is based on a minimum spanning tree approach, DPT's average over random walks (equation (1)) is significantly more robust (**Fig. 2e**). Furthermore, DPT is scalable to high cell numbers, whereas Monocle is not. In comparison to Wanderlust and Wishbone, which rely on an approximate and computationally costly sampling of shortest paths on graphs, DPT's average over random walks (equation (1)) is exact and computationally cheap. Regarding the recently published Wishbone: although Wishbone—in contrast to Wanderlust—is able to identify branching events, it does this using a number of preprocessing steps and independent algorithms (preprocessing by dimension reduction and manual selection of components in the dimension reduction). Furthermore, its pseudotime distance is based on a shortest-path distance between 'way-point' cells which, in the presence of branching, has to be computed in a rather complicated, iterative way and then no longer constitutes a good proxy for geodesic distance. By contrast, DPT consists of a single clean definition of a pseudotime measure and a simple algorithm for its computation. We believe that this fundamental difference in method design is responsible for the practical advantages of DPT over Wishbone (**Supplementary Note 7**).

Data analysis and experiments. *Detecting transcriptional changes.* To identify the succession of switch-like transcriptional changes revealed by the pseudotemporal order in qPCR data, we computed an approximate derivative of the smoothed gene expression level along branch 1. A switch-like change is defined as the maximum in the derivative (details in **Supplementary Note 2**, section 2).

Differential expression analysis. We employed a generalized linear model that allows us to quantify the proportion of cells expressing a certain gene as well as the mean expression level, a modified Hurdle model[16]. Briefly, the model has two parts: a discrete part to determine whether a gene is expressed and a continuous part using a normal distribution to quantify gene expression. Then, a likelihood ratio test is used to identify differentially expressed genes (details in **Supplementary Notes 2**, section 3 and **3**, section 4 and Finak *et al.*[16]).

First example: early blood development data (ESC qPCR). We analyzed a single-cell qPCR data set (normalized version with 3,934 cells, 42 genes) focusing on early blood development[13]. For each gene, the limit of detection (LOD) was the average Ct value for the last dilution at which all six replicates had positive amplification. The overall LOD of 25 for the gene set was the median Ct value across all genes. Raw Ct values and normalized data can be found in **Supplementary Table 7** of Moignard *et al.*[13]. Gene expression was subtracted from the LOD and normalized on a cell-wise basis to the mean expression of the four housekeeping genes (*Eif2b1*, *Mrpl19*, *Polr2a* and *Ubc*) in each cell. Cells that did not express all four housekeeping genes were excluded from subsequent analysis, as were cells for which the mean of the four housekeepers was ± 3 s.d. from the mean of all cells. A dCt value of −14 was then assigned where a gene was not detected. 85–90% of sorted cells were retained for further analysis. *Gata2* did not amplify correctly, and *HoxB3* was not expressed in any cells, so these factors have been excluded from the analysis. The analyses were done on the dCt values for all transcription factors and marker genes, but not housekeeping genes. For more details, see **Supplementary Note 2**.

Second example: InDrop differentiation data. We analyzed a single-cell RNA-seq data set using the inDrop protocol from Klein *et al.*[9]. Here, single cells along with a set of uniquely barcoded primers were captured in tiny droplets and sequenced. The capabilities of this technique were demonstrated using an undirected differentiation process of mouse embryonic stem cells upon leukemia inhibitory factor (LIF) withdrawal. The data set is publicly available under the GEO accession number GSE65525. Count data were normalized by library size and $\log_{10}$ transformed (see **Supplementary Note 3**, section 1). We corrected for cell-cycle and batch effects using scLVM (ref. 20) on 2,047 highly variable genes (see **Supplementary Table 3** in Klein *et al.*[9] and **Supplementary Data**). Hierarchical clustering was performed in R (http://www.r-project.org/) using the hclust package on quantile-normalized data (**Supplementary Note 3**, section 2) and displayed with ComplexHeatmap package, where the distance was defined as 1 − correlation between all samples (**Supplementary Note 3**, section 3). In addition, we performed a rank sums test on the first side branch that identified apoptotic genes as being significantly differently expressed as compared with the initial pluripotent and the late epiblast-like cells (**Supplementary Note 3**, section 4). For more details, see **Supplementary Note 3**.

Relation of pseudotime in snapshot data to actual time. In contrast to measurements in actual time, e.g., from time-lapse microscopy measurements, high-throughput snapshot experiments only encode the progression stage of development, but not the stochastic trajectories of single cells. We measure this progression stage using the geodesic distance on the data manifold and refer to it as 'universal time'. Here, the assumption is that the data manifold is representative for the deterministic program underlying stochastic cellular processes. For time-lapse data, universal time can be constructed by estimating the velocity $v(t)$ tangential to the manifold (C) from each single-cell trajectory. Universal time, i.e., the geodesic distance

$$s(t) = \int_{C:[s(0),s(t)]} ds = \int_0^t dt' \,|\, \mathbf{v}(t') \,| \approx \int_0^t dt' \frac{1}{\rho(t')}$$

then quantifies the arc length along the manifold, where $\tilde{n}\,(t)$ denotes the local density of samples on a single-cell trajectory (see **Supplementary Note 6**). Pseudotimes are proxies for universal time (**Supplementary Note 6**). Our proposed DPT approximates universal time better than other pseudotime schemes as it does not involve dimension reduction, and it approximates universal time better than diffusion distance[11] as it accounts for random walks on all length scales.

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